

AsCL: An Asymmetry-sensitive Contrastive Learning Method for Image-Text Retrieval with Cross-Modal Fusion

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Abstract—The image-text retrieval task aims to retrieve relevant information from a given image or text. The main challenge is to unify multimodal representation and distinguish fine-grained differences across modalities, thereby finding similar contents and filtering irrelevant contents. However, existing methods mainly focus on unified semantic representation and concept alignment for multi-modalities, while the fine-grained differences across modalities have rarely been studied before, making it difficult to solve the information asymmetry problem. In this paper, we propose a novel asymmetry-sensitive contrastive learning method. By generating corresponding positive and negative samples for different asymmetry types, our method can simultaneously ensure fine-grained semantic differentiation and unified semantic representation between multi-modalities. Additionally, a hierarchical cross-modal fusion method is proposed, which integrates global and local-level features through a multimodal attention mechanism to achieve concept alignment. Extensive experiments performed on MSCOCO and Flickr30K, demonstrate the effectiveness and superiority of our proposed method.

Index Terms—Image-text Retrieval, Information Asymmetry, Contrastive Learning, Cross-modal Fusion

I. INTRODUCTION

Image-text retrieval aims to search for relevant text based on visual queries and vice versa. Early efforts contribute significantly to the learning of unified multimodal representation and visual-textual concept alignment, but it still faces the following challenge: the existing work rarely considers the problem of cross-modal information asymmetry, which neglects the fine-grained differences between modalities.

The information asymmetry problem in the multimodal retrieval task refers to the unequal information capacity between different modalities, *i.e.*, when describing the same scene, one modality may contain more or less information than the other modality. Image modality objectively describes the scene based on pixels, while text modality describes the scene based on characters or words. It should be noted that different modalities have different emphases on the description of a certain scene. For a certain image, the paired text may be partially similar to the image and partially different from the image. The information asymmetry problem aggravates the

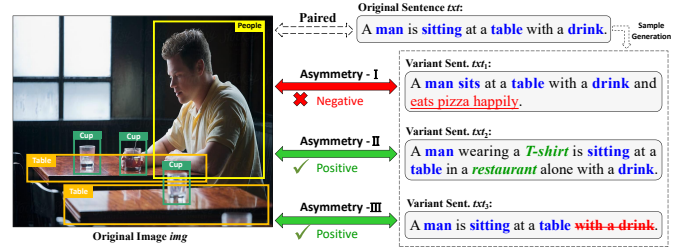


Fig. 1. Fine-grained information asymmetry types.

difficulty of the image-text retrieval task. As shown in Figure 1, for a given paired image-text pair, (img, txt) , we subdivided information asymmetry into three types. (1) Asymmetry-I: The text contains redundant information that does not exist in the image. For variant txt_1 , although txt_1 contains most of the objects corresponding to image img , such as “people”, “table”, and “cup”, txt_1 mistakenly contains “pizza” that is not mentioned in image. Therefore, txt_1 should be considered a negative of img . However, existing models would mistakenly retrieve txt_1 as a positive because it contains most of the information in the image. (2) Asymmetry-II: Compared to the original paired text, the variant text contains more relevant information belonging to the corresponding image. For variant txt_2 , it is a more informative illustration of img , which contains more details about clothing and location without irrelevant content, and should be regarded as a positive of img . (3) Asymmetry-III: Compared to the original paired text, the variant text discards partial information contained in the given image, but still conforms to the description of the image. For variant txt_3 , although the word “drink” has been deleted, it should still be considered a positive of img as it fits the description of img . However, existing models may mistakenly identify it as a negative or lower its retrieval ranking because the proportion of overlapping concepts between txt_3 and img has decreased.

To address the above issues, a novel asymmetry-sensitive

contrastive learning method was presented. For each fine-grained information asymmetry type, we generated corresponding positive or negative textual samples with both partially similar parts and subtly different parts, which can be leveraged in the optimization of contrastive learning and acquire more discriminative multimodal semantic representations. In particular, for Asymmetry-I, we generated negative samples by adding noise information into the embedding layer to fully enhance the diversity of the generated negatives. For Asymmetry-II and Asymmetry-III, we generated positive samples by concatenating and truncating keywords on the original input statements, respectively, in order to enrich the diversity of semantic descriptions for the same scenario.

Our method enhanced the sensitivity of the contrastive learning algorithm to the fine-grained information asymmetry across modalities by increasing the diversity of positive and negative samples, and provided semantic representation support for the subsequent multimodal retrieval task based on semantic similarity. In addition, in order to capture more sophisticated correlations across modalities, a hierarchical cross-modal fusion has also been proposed, for achieving multimodal concept alignment at both image-text and region-word levels, through a cross-modal attention mechanism.

The major contributions can be concluded as follows: (1) An asymmetry-sensitive contrastive learning method was proposed to solve the fine-grained information asymmetry problem between images and texts, where corresponding positives and negatives for each asymmetry type are generated to achieve unified semantic representation for better cross-modality retrieval based on semantic similarity. (2) We also presented an image-text feature fusion and semantic alignment method based on a cross-modal attention mechanism for high-quality modality interaction, from both global and local perspectives. (3) Extensive experiments verified that our method outperformed the state-of-the-art baselines.

II. OUR ASCL METHODOLOGY

A. Multi-modal Feature Representation

As shown in Figure 2, for each input image, we extracted K region features with the pretrained Faster-RCNN model and then projected them into a D -dimensional space, denoted as $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_K] \in \mathbb{R}^{K \times D}$. Meanwhile, we acquired the global feature with the pretrained ResNet152 model and then transformed it into a D -dimensional vector, i.e., $\mathbf{G} \in \mathbb{R}^D$.

For each input text, we utilized BERT to obtain word embeddings, $\mathbf{W} = [\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_L] \in \mathbb{R}^{L \times D}$, where L is the number of words and D is the dimension of word features.

B. Information Asymmetry and Sample Generation

Information asymmetry quantifies the differences between the sets of image and text information. According to fine-grained information asymmetries, we generated different samples. **Asymmetry-I**: The text contains more redundant information that does not belong to the corresponding image. We used four methods to generate diverse negative samples by adding noise information to the textual embedding layers:

(i) Gaussian Noise: Gaussian noise is random variation that follows a Gaussian (normal) distribution. We directly attached it to $\mathbf{W}$. (ii) Token Shuffling: The order of tokens in $\mathbf{W}$ was randomly altered. (iii) Token Cutoff & Feature Cutoff: We assigned zero to a row (token) or a column (feature) of $\mathbf{W}$. (iv) Dropout: We randomly discarded elements in $\mathbf{W}$ based on a specific probability. **Asymmetry-II**: The text contains more relevant information that belongs to the corresponding image. For each image in the original dataset, there are five corresponding captions sharing similar concepts. We generated a long positive sentence by randomly selecting two related sentences and concatenating them into the long sentence. **Asymmetry-III**: The text discards partial relevant information that belongs to the corresponding image. We truncated the original text to generate a short but positive sentence, which increases the diversity of positives.

C. Cross-modal Fusion

Local Region-Word Fusion. Due to the bidirectional requirement of image-text retrieval, we defined $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_M] \in \mathbb{R}^{M \times D}$ and $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_N] \in \mathbb{R}^{N \times D}$ to denote the given query (image or text) and the retrieved results (text or image), respectively, where $\mathbf{x}_i \in \mathbb{R}^D$ and $\mathbf{y}_i \in \mathbb{R}^D$ are represented as region-level or word-level features. In this work, we implemented two single symmetrical versions of formula. Hence, we set $\mathbf{X} := \mathbf{V}(M := K)$ and $\mathbf{Y} := \mathbf{W}(N := L)$ for the image-to-text (I2T) version, and $\mathbf{X} := \mathbf{W}(M := L)$ and $\mathbf{Y} := \mathbf{V}(N := K)$ for the text-to-image (T2I) version. “:=” is an assignment operator. $\mathbf{V} \in \mathbb{R}^{K \times D}$, $\mathbf{W} \in \mathbb{R}^{L \times D}$ are defined in Section 2.1, and will be assigned to $\mathbf{X}$ or $\mathbf{Y}$ in different retrieval versions.

Instead of treating all region-word pairs equally, a multi-head cross-modal attention mechanism was adopted to allocate weights to regions and words based on their contribution.

$$\begin{aligned} \mathbf{Y}^* &= \text{MultiHead}(\text{Query} := \mathbf{X}, \text{Key} := \mathbf{Y}, \text{Value} := \mathbf{Y}) \\ &= \text{Concat}(\text{head}_1, \dots, \text{head}_i, \dots, \text{head}_H) \mathbf{Z}^O \end{aligned} \quad (1)$$

where $\mathbf{Y}^* \in \mathbb{R}^{M \times D}$, $\mathbf{Z}^O \in \mathbb{R}^{D \times D}$, H is the number of attention heads, and $\text{Concat}(\cdot)$ represents the concatenation operation across the feature dimension D . In this work, $\text{head}_i = \text{Att}(\text{Query}_i := \mathbf{X} \mathbf{Z}_i^X, \text{Key}_i := \mathbf{Y} \mathbf{Z}_i^Y, \text{Value}_i := \mathbf{Y} \mathbf{Z}_i^Y)$, where Att refers to the scaled dot-product attention, and $\mathbf{Z}_i^X, \mathbf{Z}_i^Y \in \mathbb{R}^{D \times \frac{D}{H}}$.

For I2T, the output $\mathbf{Y}^*$ denotes the region-attended text representations $\mathbf{W}^* = [\mathbf{w}_1^*, \mathbf{w}_2^*, \dots, \mathbf{w}_K^*] \in \mathbb{R}^{K \times D}$ for each region, while for T2I, it refers to the word-attended image representations $\mathbf{V}^* = [\mathbf{v}_1^*, \mathbf{v}_2^*, \dots, \mathbf{v}_L^*] \in \mathbb{R}^{L \times D}$ for each word. Here, $\mathbf{w}_i^* \in \mathbb{R}^D$ represents the region-attended text representation for the i -th region, and $\mathbf{v}_i^* \in \mathbb{R}^D$ represents the word-attended image representation for the i -th word.

Global Image-Text Fusion. To further capture complex cross-modal correlation, we jointly mapped holistic images and sentences into a common space for global semantic consistence and heterogeneity minimization. To be specific, for the region-attended text representations $\mathbf{W}^*$ obtained in

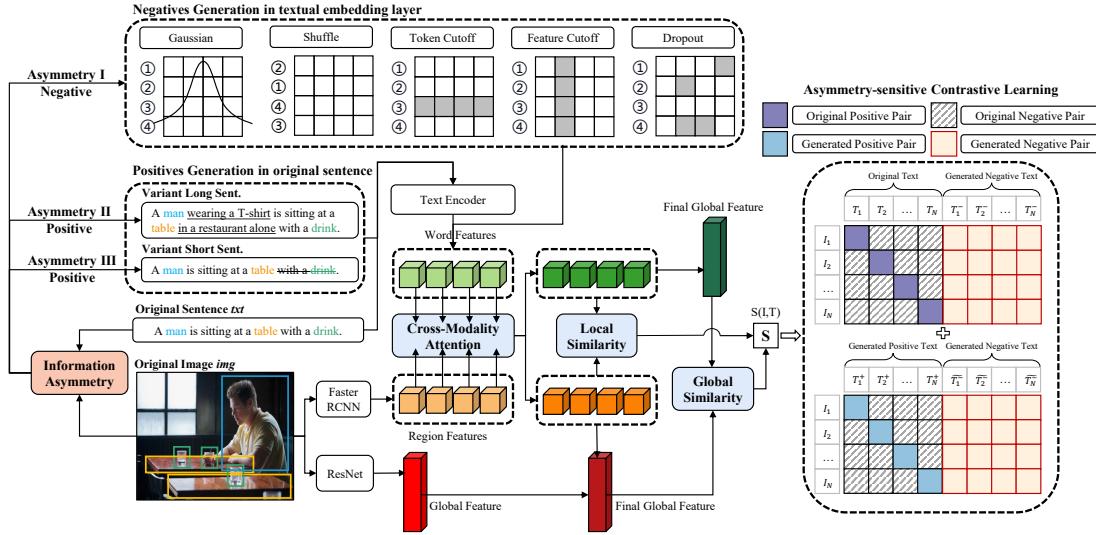


Fig. 2. The overview of our proposed AsCL Method.

the process of local fusion, we first computed its average vector (denoted as $\bar{\mathbf{W}}^* = \frac{\sum_{i=0}^K \mathbf{w}_i^*}{K}$), and then projected it into a common space. The output result is $\mathbf{W}_g = \mathbf{X}_w^T \bar{\mathbf{W}}^*$, which serves as the final global representation of text, and $\mathbf{X}_w$ is a learnable embedding matrix. Similarly, we separately projected the average vector of $\mathbf{V}^*$ (represented as $\bar{\mathbf{V}}^* = \frac{\sum_{i=0}^L \mathbf{v}_i^*}{L}$) and the feature vector of the entire image (represented as $\mathbf{G}$) into the common embedding space as $\mathbf{V}_{g1} = \mathbf{X}_v^T \bar{\mathbf{V}}^*$ and $\mathbf{V}_{g2} = \mathbf{X}_g^T \mathbf{G}$. $\mathbf{X}_v$ and $\mathbf{X}_g$ are all learnable embedding matrixes. The final global representation of image can be fused as $\mathbf{V}_g = \text{Fusion}(\mathbf{V}_{g1}, \mathbf{V}_{g2})$.

D. AsCL for I-T Matching

The matching score between image I and text T consists of two components: local similarity score and global similarity score. According to the region-attended text representations $\mathbf{W}^*$ and the word-attended image representations $\mathbf{V}^*$, the local matching score is formulated as,

$$S_{local}(I, T) = \frac{\sum_{i=1}^k R(\mathbf{v}_i, \mathbf{w}_i^*)}{2K} + \frac{\sum_{j=1}^L R(\mathbf{w}_j, \mathbf{v}_j^*)}{2L} \quad (2)$$

, where $R(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x}^T \mathbf{y}}{\|\mathbf{x}\| \cdot \|\mathbf{y}\|}$. Meanwhile, the global matching score between image I and text T is represented as:

$$S_{global}(I, T) = R(\mathbf{V}_g, \mathbf{W}_g). \quad (3)$$

Based on the local and global matching scores, the similarity score between image I and text T can be defined as:

$$S(I, T) = u_1 \cdot S_{local}(I, T) + (1 - u_1) \cdot S_{global}(I, T) \quad (4)$$

where u_1 is a hyper-parameter.

On this basis, we proposed a novel asymmetry-sensitive contrastive learning method by exploiting the generated positives and negatives according to our defined asymmetry types.

I-T Matching for Asymmetry-I. During training, there are N image-text pairs in a batch. According to our defined

Asymmetry-I, we generated N negative sentences by adding noise. For each positive pair (I, T) , we retrieved $N-1$ in-batch negative images $\{\hat{I}_n\}_{n=1}^{N-1}$, $N-1$ in-batch negative sentences $\{\hat{T}_n\}_{n=1}^{N-1}$, and especially N generated negative sentences $\{T_n^-\}_{n=1}^N$ for Asymmetry-I. Therefore, the objective function for Asymmetry-I can be formulated as follows:

$$L_I(I, T) = \frac{e^{S(I, T)/\tau}}{e^{S(I, T)/\tau} + \sum_{n=1}^{N-1} e^{S(\hat{I}_n, T)/\tau} + \sum_{n=1}^N e^{S(I, \hat{T}_n)/\tau}} + \frac{e^{S(I, T)/\tau}}{e^{S(I, T)/\tau} + \sum_{n=1}^{N-1} e^{S(I, \hat{T}_n)/\tau} + \sum_{n=1}^N \alpha_n \cdot e^{S(I, T_n^-)/\tau}} \quad (5)$$

where α_n is set to zero if the similarity score of the image-text pair exceeds the positive pair, otherwise we set α_n to 1.

I-T Matching for Asymmetry-II&III. For each positive image-text pair (I, T) , we generated the content-variant long sentence or short sentence as another positive sample, T^+ , according to our defined Asymmetry-II and Asymmetry-III. Here, (I, T^+) is a newly generated positive pair, while T^+ is also considered as a negative sample for other images in the same batch. Likewise, we added noise information as additional negative samples. Consequently, there exist $N-1$ in-batch negative images $\{\hat{I}_n\}_{n=1}^{N-1}$, $N-1$ in-batch negative sentences $\{\hat{T}_n\}_{n=1}^{N-1}$, and N generated negative sentences $\{T_n^-\}_{n=1}^N$ towards every positive pair (I, T^+) . For Asymmetry-II & III, T^+ plays a similar role in Equation 6 as T in Equation 5.

$$L_{II\&III}(I, T^+) = \frac{e^{S(I, T^+)/\tau}}{e^{S(I, T^+)/\tau} + \sum_{n=1}^{N-1} e^{S(\hat{I}_n, T^+)/\tau} + \sum_{n=1}^N e^{S(I, \hat{T}_n)/\tau}} + \frac{e^{S(I, T^+)/\tau}}{e^{S(I, T^+)/\tau} + \sum_{n=1}^{N-1} e^{S(I, \hat{T}_n)/\tau} + \sum_{n=1}^N \alpha_n \cdot e^{S(I, T_n^-)/\tau}} \quad (6)$$